

Stage 4: Sprint Plan & Task Backlog

Stage Four: Sprint Plan

Forward Deployed AI Engineering — Capstone Project

Deciding who does what, in what order, and what finished means. A plan is not a list of tasks. It is a statement of sequence, ownership and completion that the team has actually agreed to.

Week Two · Day One (updated Sep 19 with completed engineering backlog)

How to Plan Three Weeks of Work

You have already lost a third of the project to discovery and requirements, which is as it should be. What remains has to be sequenced carefully, because several parts of the build depend on others and the dependencies are not always obvious until you hit them.

Three Rules That Make the Difference

1. Every item has an estimate in hours and a priority. When you run short of time, and you will, the priority column is what decides for you rather than whatever felt urgent that morning.
2. Every item has a definition of done written before work starts. Written afterwards, it describes whatever happened to be produced.
3. The retrieval layer and the evaluation harness come early, not late. Almost everything else depends on one or the other, and leaving the harness until the end means discovering your results too late to act on them.
4. The full unattended run over the hidden evaluation set happens in week two, not week three. It always exposes problems that never appear when you are testing one ticket at a time.

On working alone: Working individually removes the coordination overhead and replaces it with a different problem: nobody will notice when you have been stuck for six hours. Set yourself a rule that anything blocking you for more than an hour gets written down and either escalated or worked around, and check that list at the end of each day.

Section 1: Your Capacity

Before you plan anything, work out honestly how much time you actually have.

Week	Hours realistically available	Other commitments that week	Days you cannot work
Week 1 (Sep 7–12)	40h	None	Sep 6 (Sun, pre-start)
Week 2 (Sep 13–18)	50h	None	—
Final push (Sep 19–20)	18h	Submission deadline by end of day Sep 20	Sep 21+ out of scope

Note on this timeline: This is a compressed 2-week sprint (Sep 6–20). All substantive software engineering, testing, containerization, and CI pipeline automation landed in Weeks 1–2 (Sep 7–19), leaving Sep 20 for final presentation recording and submission packaging.

Section 2: The Backlog

List every item of work. Estimate in hours rather than in vague sizes, because hours can be compared against the capacity you recorded above. Mark dependencies explicitly, because the order matters more when there is only one of you.

ID	Item	Est. hours	Priority	Depends on	Status	Definition of done
B-01	Environment and dependency setup	3h	P0	—	Done (Sep 7)	<code>uv pip install</code> runs clean; <code>.env</code> populated with all required keys; <code>python -c "import openai, chromadb"</code> exits 0; satisfies A1 pre-condition

ID	Item	Est. hours	Priority	Depends on	Status	Definition of done
B-02	Data loading and normalisation	4h	P0	B-01	Done (Sep 8)	<code>development_tickets.json</code> (500) and <code>ground_truth_responses.json</code> (200) load without error; all four channels (email, live-chat, docs-comment, forum) represented and normalised; satisfies A2
B-03	Document chunking and embedding	6h	P0	B-02	Done (Sep 9)	<code>ingest.py</code> completes without error; ChromaDB collection populated; chunk size and overlap documented in report; satisfies A4 pre-condition
B-04	Vector store and retrieval (ChromaDB startup index rebuild)	4h	P0	B-03	Done (Sep 9)	<code>retrieve.py</code> returns top-K passages with scores for an arbitrary query; returned doc IDs resolve to real corpus entries; index rebuilt on startup to avoid stale docs; satisfies A4
B-05	Evaluation harnesses (all 6 agents)	12h	P0	B-02	Done (Sep 12)	All 6 <code>run*_eval.py</code> scripts run to CSV output; three-tier API key priority (Groq > Gemini > OpenRouter); <code>--sample</code> , <code>--experiment-name</code> , <code>--concurrency</code> flags working
B-06	Intent and urgency classifiers (PR-01, PR-03)	8h	P0	B-02	Done (Sep 11)	PR-01 <code>escalation_safety</code> ≥ 0.95 ; PR-03 <code>urgency_safety</code> ≥ 0.90 and <code>high_urgency_recall</code> ≥ 0.90 ; confidence is numeric 0–1; satisfies A3
B-07	Coordinator routing logic (PR-00)	8h	P0	B-06	Done (Sep 12)	FLAG A / EXIT B architecture implemented; same ticket produces same decision (deterministic); <code>routing_accuracy</code> ≥ 0.90 ; escalations carry draft + sources; satisfies A5
B-08	Answer generation with citations (PR-05)	6h	P0	B-04	Done (Sep 11)	<code>must_not_claim_rate</code> = 0.0; every claim cites a doc ID that resolves to a real retrieved passage; "I don't know" returned

ID	Item	Est. hours	Priority	Depends on	Status	Definition of done
						when retrieval hits nothing; satisfies A6
B-09	Guardrails and validation (PR-guardrails)	4h	P0	B-08	Done (Sep 12)	PII detector and prompt-leak check run on every response before release; at least one guardrail blocks a crafted trigger ticket; activations written to log; satisfies A7
B-10	Decision logging	3h	P0	B-07	Done (Sep 12)	Every automated decision written to <code>decisions.db</code> with ticket_id, decision, intent, urgency, confidence, timestamp, and reason; count reconciles against tickets processed; satisfies A8
B-11	Full unattended run — THE GATE (A9 + A10)	4h	P0	B-05, B-09	Done (Sep 16)	80 validation tickets processed in single unattended run; 0 errors; <code>metrics_20260916T175514Z.json</code> auto-produced covering all four groups (volume, business, technical, governance); satisfies A9 and A10
B-12	Monitoring and dashboards	6h	P3	B-10	Cut (Deferred)	Operational visibility satisfied via metrics JSON auto-generation & SQLite decision store; live Grafana dashboard deferred
B-13	Automated GitHub Actions CI pipeline	4h	P1	B-18	Done (Sep 19)	<code>.github/workflows/ci.yml</code> runs unit tests, enforces ≥ 90% coverage gate, tests Docker container build, and triggers automatically on push/PR to <code>main</code> ; satisfies A12
B-14	Fairness audit	3h	P2	B-11	Done (Sep 19)	Metrics sliced by <code>customer_tier</code> and <code>channel</code> ; demographic disparity in urgency classification analyzed and documented in governance section
B-15	Report, video and submission	10h	P0	all	In Progress	4-part submission complete: video (18–22 min MP4), 20–30 page

ID	Item	Est. hours	Priority	Depends on	Status	Definition of done
	package				(Sep 20)	report PDF, 5 Stage workbooks + effort log, source code repo passing clean checkout
B-16	Prompt iteration and tuning & version history	10h	P1	B-05	Done (Sep 15)	Urgency v3.5 and Coordinator v1.1 merged; version history captured in <code>src/prompts/</code> with change notes; active prompt versions documented
B-17	README runbook + <code>.env.example</code> + Docker dual execution	4h	P0	B-01	Done (Sep 18)	<code>README.md</code> details native & Docker execution; <code>docker-entrypoint.sh</code> runs one-shot batch + background REST API; <code>.env.example</code> populated; satisfies A1
B-18	Unit test suite expansion & coverage gate ($\geq 90\%$)	5h	P0	B-11	Done (Sep 19)	Expanded from 33 to 51 unit tests (100% pass rate); achieved 91.47% test coverage (<code>pytest --cov=src --cov-fail-under=90</code>); satisfies A12
B-19	Harness parametrisation & REST API batch endpoint	3h	P0	B-11	Done (Sep 18)	<code>POST /tickets/batch</code> endpoint implemented; harness accepts <code>--input</code> and <code>--output</code> flags; satisfies A9 critical requirement
B-20	Metrics report interpretation document	2h	P0	B-11	Done (Sep 19)	Cites <code>metrics_20260916T175514Z.json</code> ; all four metric groups interpreted; PRD compliance verified
B-21	Guardrails AI Hub key resolution & <code>requirements.txt</code> pinning	2h	P1	B-09, B-13	Done (Sep 19)	Audited <code>guardrailsai.com/hub/keys</code> ; removed obsolete <code>GUARDRAILS_API_KEY</code> token check; pinned exact versions in <code>requirements.txt</code>

Section 3: The Week Two Plan in Detail

Week 2 = Sep 13 (Sun) – Sep 18 (Fri).

Day	What will be finished by the end of the day	Hours	Risk to this
Sunday Sep 13	Week 2 planning & coordinator architecture review complete; exit strategy & fallback mechanism defined	6h	Architecture ambiguity (mitigated: decision logged in Stage 3 Prompt Library)
Monday Sep 14	Coordinator FLAG A / EXIT B architecture confirmed correct; root cause of <code>load_dotenv</code> shell-override bug found and fixed (<code>override=True</code>); coordinator eval harness running without errors on 5-ticket smoke test	10h	Shell environment variables overriding <code>.env</code> (fixed with <code>override=True</code>)
Tuesday Sep 15	Three-tier API key support (Groq > Gemini > OpenRouter) applied to all 6 eval harnesses and <code>src/agents/_llm.py</code> ; urgency v3.3–v3.5 tuning completed; coordinator v1.1 merged; versioned prompt library updated	10h	Groq model list gaps; Gemini free-tier 429 rate limits (mitigated: Groq fallback)
Wednesday Sep 16	Urgency v3.5 and coordinator v1.1 merged; THE GATE run: 80 validation tickets, 0 errors, <code>metrics_20260916T175514Z.json</code> auto-saved; A9 and A10 satisfied	10h	Model API outage or rate limits; mitigated by three-tier fallback
Thursday Sep 17	Stage 3 Prompt Library updated (urgency v3.5, coordinator v1.1, SYS-00 CLAUDE.md added); Sprint Plan updated; B-19 harness <code>--input</code> / <code>--output</code> parametrisation started	8h	Documentation scope creep
Friday Sep 18	B-17: README + <code>.env.example</code> verified; Dockerfile + <code>docker-compose.yml</code> + <code>docker-entrypoint.sh</code> created for simultaneous batch & REST API execution; <code>POST /tickets/batch</code> endpoint added to <code>src/api.py</code>	8h	Port conflicts or container entrypoint issues (resolved with shell signal trapping)

Section 4: The Final Push Plan in Detail (Week 3)

Final Push = Sep 19 (Sat) + Sep 20 (Sun, deadline).

Day	What will be finished by the end of the day	Hours	Risk to this
Saturday Sep 19	Unit test suite expanded from 33 to 51 tests with 91.47% test coverage (<code>pytest --cov=src --cov-fail-under=90</code>); GitHub Actions CI pipeline <code>.github/workflows/ci.yml</code> implemented; Guardrails AI Hub key requirement audited & removed; <code>requirements.txt</code> dependencies fully pinned; master <code>checklist.md</code> created	10h	CI pipeline failure on remote runners (mitigated: verified local docker build & pytest gate)
Sunday Sep 20 — DEADLINE	Effort Log PDF completed; 20–30 page report PDF compiled; video recorded & rendered (18–22 min MP4); 4-part submission package <code>AyushYadav_Capstone_Submission.zip</code> assembled and uploaded	8h	Video encoding delays or submission portal congestion (target upload by 18:00)

Section 5: What You Will Drop If You Run Out of Time

Deciding beforehand what gets cut protects the parts of the build that carry the most marks.

Item	Cut order	Consequence of cutting it	What you will say about it in the report
B-12 — Live monitoring dashboard	1st to go	No live Grafana/Prometheus dashboard; visibility satisfied via metrics JSON and SQLite decision log	"A live monitoring dashboard was scoped but deferred as it is not required by A1–A12. Guardrail activation counts and latency percentiles are captured in the metrics report and decision log, which satisfies the governance requirement."
B-14 — Detailed demographic slice audit	2nd to go	Sliced analysis limited to tier × channel aggregate tables	"A detailed demographic slice audit (customer tier × channel) was planned. Aggregate urgency accuracy was verified to be consistent with label distributions. Sliced analysis is identified as immediate follow-on work."

Section 6: The Daily Check-In Record

Ten minutes at the start of each day, recording progress and blockers.

Date	Finished since yesterday	Doing today	Blocked by
Sun Sep 7	Project brief, stage structure, and submission guide read; repository scaffolded	Environment setup (B-01); <code>.env</code> ; package install	None
Tue Sep 9	B-01 done; packages installed; <code>.env</code> working	Chunking + ChromaDB (B-03, B-04); <code>ingest.py</code>	ChromaDB API — resolved with docs
Thu Sep 11	B-03, B-04 done; retrieval returning correct passages	Intent + urgency classifier prompts (B-06); draft agent (B-08)	None
Sat Sep 13	B-02, B-06, B-07, B-08, B-09, B-10 done; eval harnesses written (B-05)	Week 2 planning; coordinator arch review	EXIT A vs FLAG A ambiguity — resolved Mon
Mon Sep 14	Week 2 plan written	Coordinator FLAG A fix; <code>load_dotenv(override=True)</code> bug fix; 5-ticket smoke test	Shell env overriding <code>.env</code> — fixed with <code>override=True</code>
Tue Sep 15	Coordinator harness running clean; FLAG A / EXIT B confirmed	Three-tier API key support all 6 harnesses; urgency v3.3–v3.5 tuning	Gemini 429 rate limits; Groq model availability (resolved: switched to <code>openai/gpt-oss-120b</code>)
Wed Sep 16	Urgency v3.5 merged; all 6 harnesses updated	THE GATE — 80-ticket unattended run; <code>metrics_20260916T175514Z.json</code>	OpenRouter key balance (mitigated: used Groq fallback)
Thu Sep 17	THE GATE complete (80/80, 0 errors); Stage 3 Prompt Library updated	Sprint Plan updated; B-19 harness parametrisation; README draft	None
Fri Sep 18	B-17 README + <code>.env.example</code> verified; Docker container + <code>docker-entrypoint.sh</code> dual execution configured	<code>P0ST /tickets/batch</code> REST API endpoint; Docker testing	Port mapping conflicts — resolved in compose config
Sat Sep 19	Test suite expanded to 51 tests with 91.47% coverage; CI workflow <code>.github/workflows/ci.yml</code> built; Guardrails AI Hub key audit complete; <code>requirements.txt</code> pinned	Sprint plan refinement; Effort log completion; Master <code>checklist.md</code> creation	None

Date	Finished since yesterday	Doing today	Blocked by
Sun Sep 20	All 5 Stage Workbooks complete; master checklist audited	Report PDF, video presentation, effort log export & final zip packaging	None

Related documents: [01 Discovery Workbook.pdf](#) · [02 Requirements PRD.pdf](#) · [03 Prompt Library.pdf](#) · [05 PRD Revision Log.pdf](#) · [AyushYadav Effort Log.pdf](#)